

ECON484 project

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Predicting Employment Outcomes among Young Adults: Education, Well-being, and Machine Learning Evidence from PSID-TAS 2023.

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.2
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.5.2
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 5.0
```

```
library(randomForest)
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
```

```
## Attaching package: 'randomForest'
```

```
## The following object is masked from 'package:ggplot2':
```

```
##
```

```
##     margin
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
##     combine
```

```
library(rpart)
```

```
## Warning: package 'rpart' was built under R version 4.5.2
```

```
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```
library(gbm)
```

```
## Warning: package 'gbm' was built under R version 4.5.2
```

```
## Loaded gbm 2.2.3
```

```
## This version of gbm is no longer under development. Consider transitioning to gbm3, https://github.com
```

```
library(nnet)
```

```
data <- read_excel("73datas.xlsx")
```

```
dim(data)
```

```
## [1] 2434 73
```

```
names(data)
```

```
## [1] "TAS"      "TAS23"    "ER30001"  "ER30002"  "ER32000"  "ER32006"
## [7] "ER35101"  "ER35102"  "ER35103"  "TA230001" "TA230050" "TA230051"
## [13] "TA230057" "TA230058" "TA230059" "TA230060" "TA230061" "TA230062"
## [19] "TA230063" "TA230065" "TA230066" "TA230067" "TA230069" "TA230070"
## [25] "TA230072" "TA230074" "TA230113" "TA230171" "TA230239" "TA230240"
## [31] "TA230241" "TA230244" "TA230427" "TA230428" "TA230628" "TA230629"
## [37] "TA230636" "TA230685" "TA230691" "TA230694" "TA230707" "TA230715"
## [43] "TA230716" "TA230861" "TA230948" "TA230950" "TA230953" "TA230954"
## [49] "TA230955" "TA230956" "TA230986" "TA230988" "TA230990" "TA230991"
## [55] "TA231000" "TA231001" "TA231025" "TA231145" "TA231146" "TA231147"
## [61] "TA231229" "TA231234" "TA232095" "TA232303" "TA232320" "TA232321"
## [67] "TA232322" "TA232325" "TA232335" "TA232392" "TA232393" "TA232394"
## [73] "TA232396"
```

```
project_data <- data %>%
  mutate(
    employed = ifelse(
      TA230239 %in% c(1,2),
      1,
      0
    )
  )
table(project_data$employed)
```

```
##
##      0      1
## 1043 1391
```

Wellbeing

```
#TA230061 happy
#TA230062 interest in life
#TA230063 satisfied
#TA230065 community
#TA230066 society better
#TA230067 people good
#TA230069 manage responsibilities
#TA230070 trusting relationships
#TA230072 confidence
#TA230074 life purpose

wellbeing_vars <- c(
  "TA230061",
  "TA230062",
  "TA230063",
  "TA230065",
  "TA230066",
  "TA230067",
  "TA230069",
  "TA230070",
  "TA230072",
  "TA230074"
```

```

"TA230074"
)

project_data <- project_data %>%
  mutate(
    across(
      all_of(wellbeing_vars),
      ~ifelse(. == 9, NA, .)
    )
  )

project_data <- project_data %>%
  rowwise() %>%
  mutate(
    wellbeing_score =
      mean(
        c_across(all_of(wellbeing_vars)),
        na.rm = TRUE
      )
  ) %>%
  ungroup()
summary(project_data$wellbeing_score)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##      1.000   3.400   4.100   4.062   4.900   6.000         4

```

```

ggplot(
  project_data,
  aes(x = wellbeing_score)
) +
  geom_histogram(
    bins = 20
  ) +
  labs(
    title = "Distribution of Well-being Scores"
  )

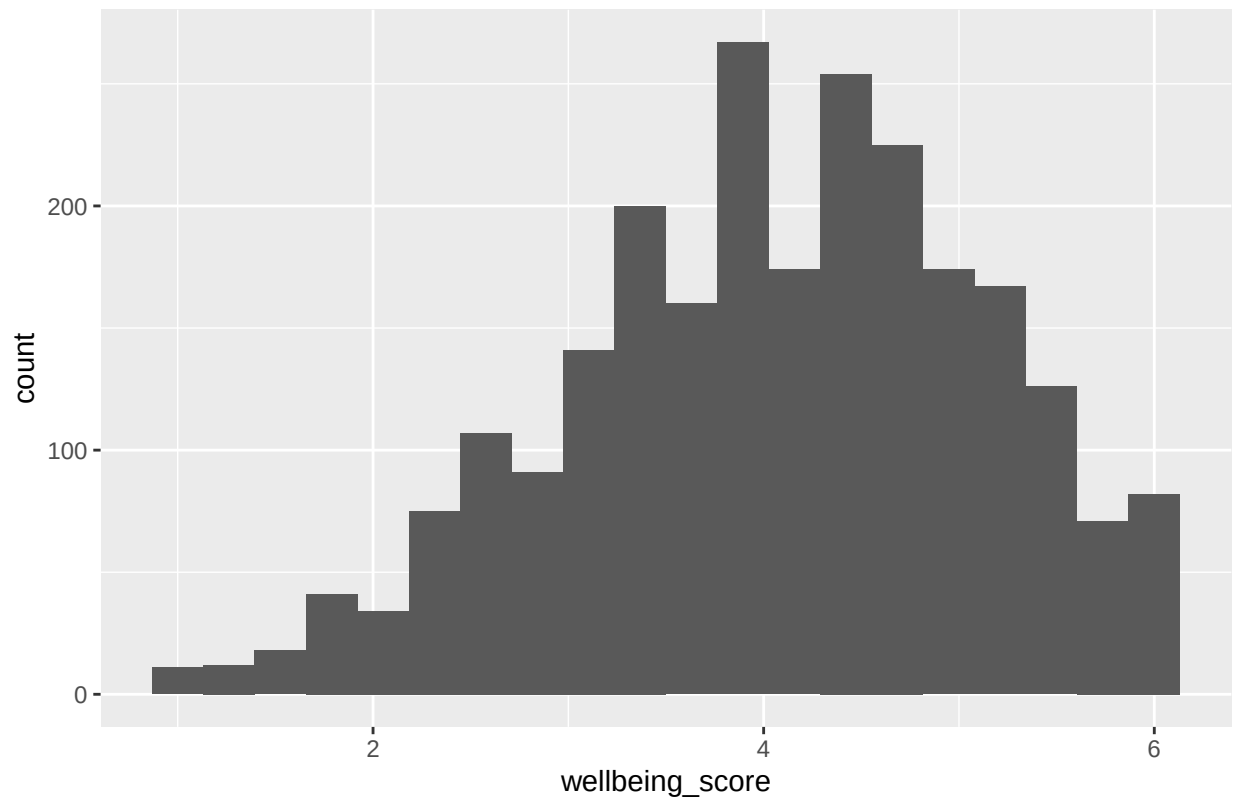
```

```

## Warning: Removed 4 rows containing non-finite outside the scale range
## ('stat_bin()').

```

Distribution of Well-being Scores



```
project_data %>%
  group_by(employed) %>%
  summarise(
    avg_wellbeing =
      mean(wellbeing_score,
        na.rm = TRUE)
  )
```

```
## # A tibble: 2 x 2
##   employed avg_wellbeing
##   <dbl>      <dbl>
## 1      0      4.03
## 2      1      4.09
```

```
#Employment vs Well-being
ggplot(
  project_data,
  aes(
    x = factor(employed),
    y = wellbeing_score
  )
) +
  geom_boxplot() +
  labs(
    x = "Employment Status",
```

```
y = "Well-being Score"
)
```

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```
#Logistic Regression
#Does well-being predict employment?
wellbeing_logit <- glm(
  employed ~ wellbeing_score,
  data = project_data,
  family = binomial
)

summary(wellbeing_logit)
```

```
##
## Call:
## glm(formula = employed ~ wellbeing_score, family = binomial,
##      data = project_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.09302    0.16184   0.575   0.565
```

```
## wellbeing_score 0.04898    0.03862    1.268    0.205
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 3317.5  on 2429  degrees of freedom
## Residual deviance: 3315.9  on 2428  degrees of freedom
## (4 observations deleted due to missingness)
## AIC: 3319.9
##
## Number of Fisher Scoring iterations: 4
```

Higher well-being is associated with a slightly higher probability of being employed, but this relationship is weak and not statistically significant in the simple logit model.

```
#Tree
set.seed(123)

wellbeing_tree_data <- project_data %>%
  select(employed, all_of(wellbeing_vars)) %>%
  na.omit()

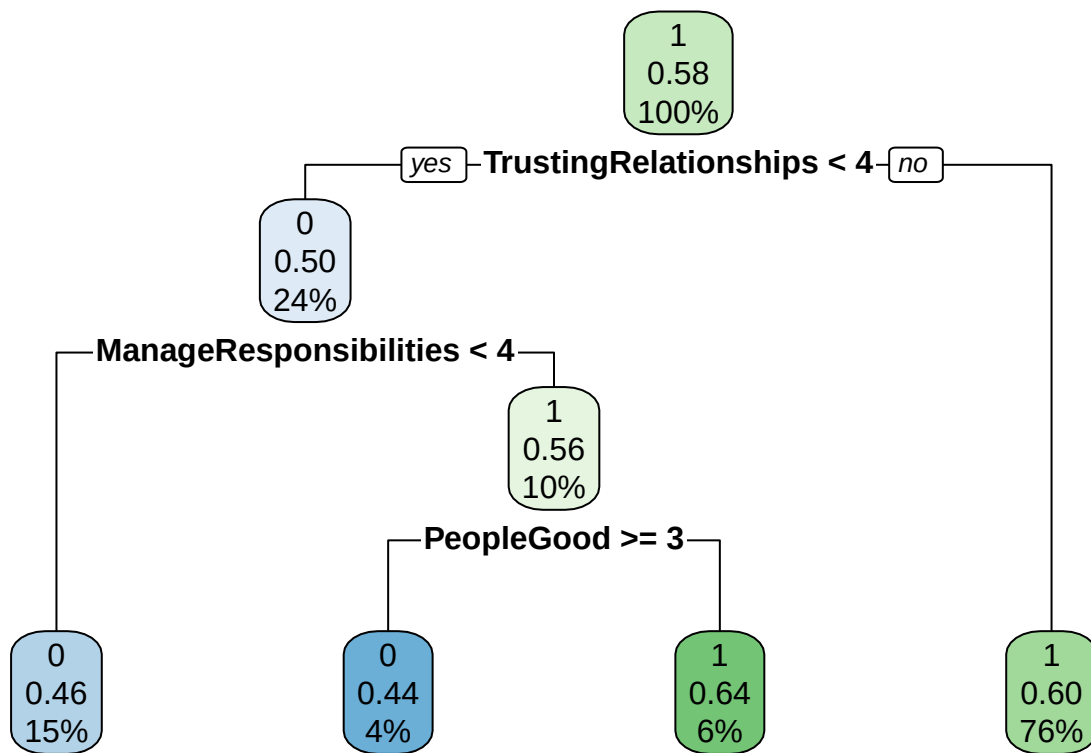
colnames(wellbeing_tree_data) <- c(
  "employed", "Happy", "InterestInLife", "Satisfied", "Community",
  "SocietyBetter", "PeopleGood", "ManageResponsibilities",
  "TrustingRelationships", "Confidence", "LifePurpose"
)

# Train/test split
n <- nrow(wellbeing_tree_data)
train_idx <- sample(seq_len(n), size = 0.7 * n)

wb_train <- wellbeing_tree_data[train_idx, ]
wb_test  <- wellbeing_tree_data[-train_idx, ]

# Fit tree on training data
wellbeing_tree <- rpart(
  employed ~ .,
  data = wb_train,
  method = "class"
)

rpart.plot(wellbeing_tree)
```



```
# Predict on test data
wb_pred <- predict(wellbeing_tree, wb_test, type = "class")
wb_test_acc <- mean(wb_pred == wb_test$employed)
wb_test_acc
```

```
## [1] 0.550765
```

The CART model failed to identify meaningful splits, suggesting that individual well-being indicators alone do not strongly distinguish employed from non-employed young adults.

```
#Random Forest
set.seed(123)

wellbeing_rf <- randomForest(
  as.factor(employed) ~ .,
  data = wellbeing_tree_data,
  ntree = 500,
  importance = TRUE
)

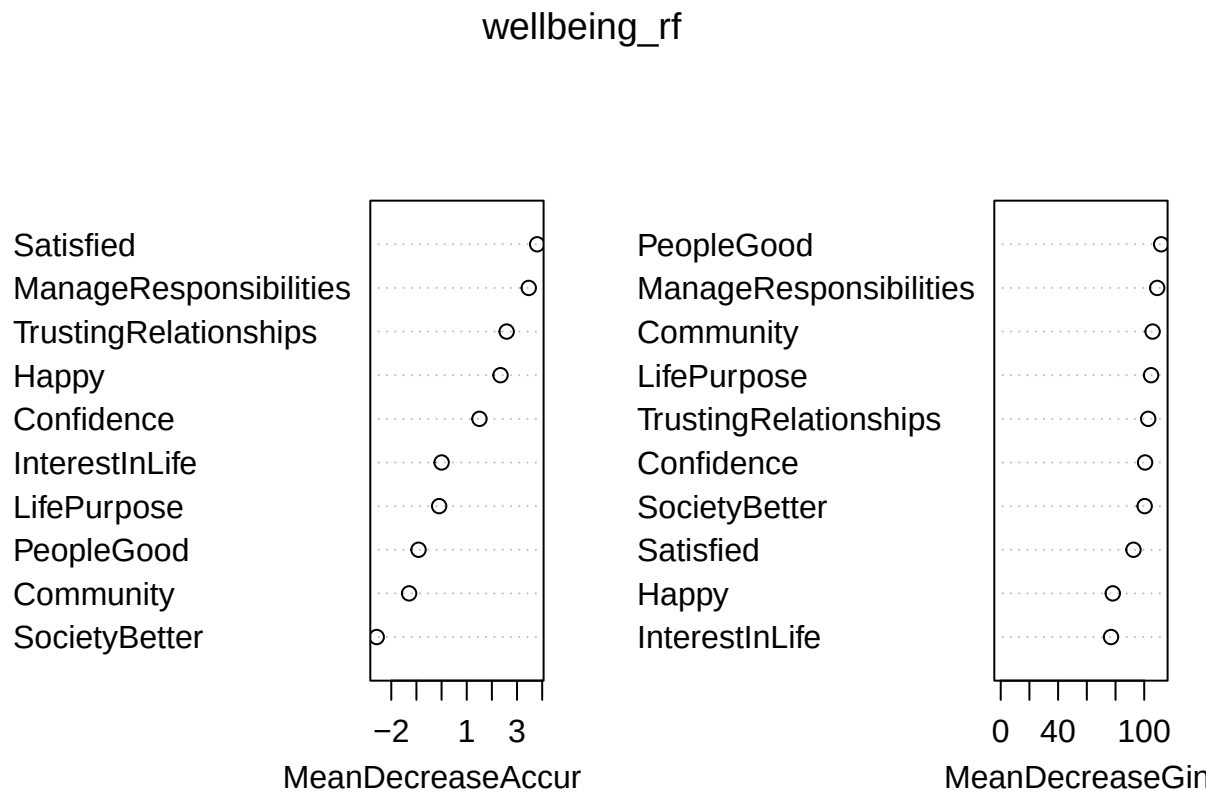
wellbeing_rf
```

```
##
## Call:
## randomForest(formula = as.factor(employed) ~ ., data = wellbeing_tree_data, ntree = 500, impor
```



```
##           Type of random forest: classification
##           Number of trees: 500
## No. of variables tried at each split: 3
##
##           OOB estimate of  error rate: 48.16%
## Confusion matrix:
##      0   1 class.error
## 0 302 723   0.7053659
## 1 431 940   0.3143691
```

```
varImpPlot(wellbeing_rf)
```



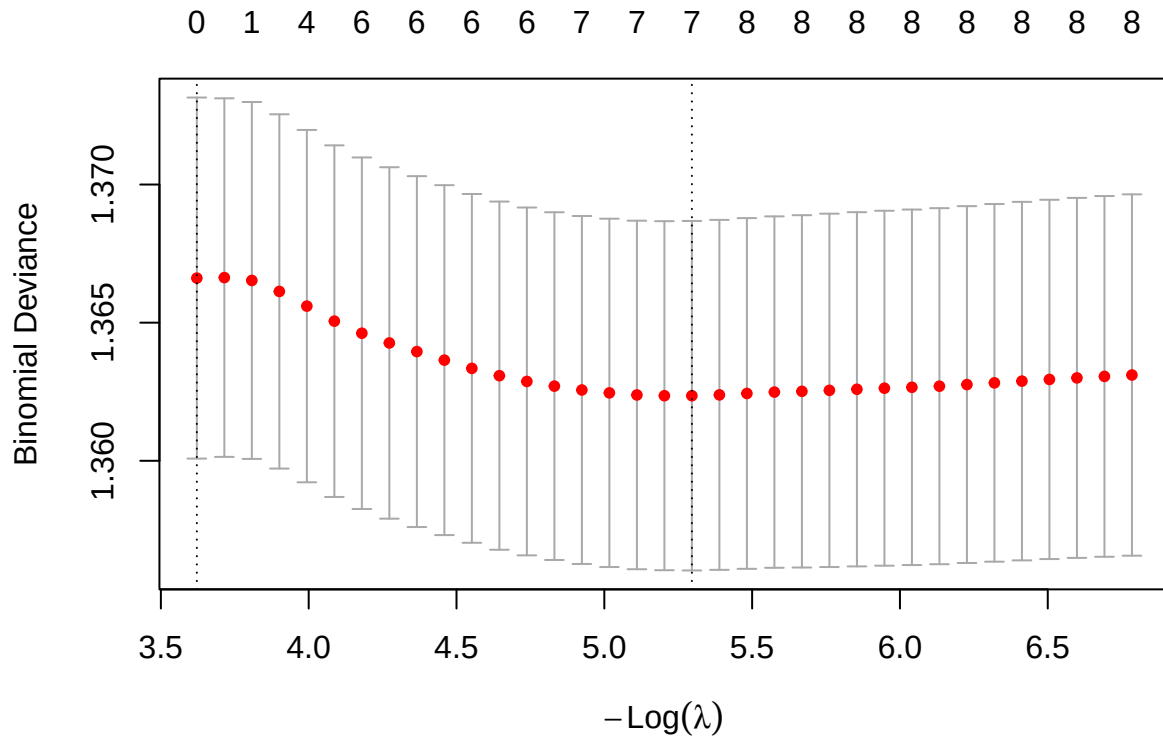
Some well-being indicators appear moderately important, but predictive power remains weak.

```
# LASSO
x_wellbeing <- model.matrix(employed ~ ., wellbeing_tree_data)[, -1]
y_wellbeing <- wellbeing_tree_data$employed

set.seed(123)

wellbeing_lasso <- cv.glmnet(
  x_wellbeing,
  y_wellbeing,
  family = "binomial",
  alpha = 1
)
```

```
plot(wellbeing_lasso)
```



```
coef(wellbeing_lasso, s = "lambda.min")
```

```
## 11 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.min
## (Intercept)   -0.102676366
## Happy          0.030246278
## InterestInLife 0.032700109
## Satisfied      0.060432021
## Community      -0.032519379
## SocietyBetter  -0.069577728
## PeopleGood     .
## ManageResponsibilities 0.024485600
## TrustingRelationships 0.006561671
## Confidence     .
## LifePurpose    .
```

```
coef(wellbeing_lasso, s = "lambda.1se")
```

```
## 11 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.1se
## (Intercept)   0.2908478
```

```
## Happy .
## InterestInLife .
## Satisfied .
## Community .
## SocietyBetter .
## PeopleGood .
## ManageResponsibilities .
## TrustingRelationships .
## Confidence .
## LifePurpose .
```

lambda.1se selected no predictors

Education

```
education_vars <- c(
  "TA232393",
  "TA230990",
  "TA231000",
  "TA231001",
  "TA230986",
  "TA230988",
  "TA230685",
  "TA232394",
  "TA232396"
)

education_labels <- c(
  "HighestEducation",
  "GradeCompleted",
  "AttendedCollege",
  "CurrentCollege",
  "EducationAspiration",
  "EducationExpectation",
  "EducationHours",
  "MotherEducation",
  "FatherEducation"
)

education_tree_data <- project_data %>%
  select(
    employed,
    all_of(education_vars)
  ) %>%
  na.omit()

colnames(education_tree_data) <- c(
  "employed",
  education_labels
)
```

```

# Logistic Regression
education_logit <- glm(
  employed ~ .,
  data = education_tree_data,
  family = binomial
)

summary(education_logit)

##
## Call:
## glm(formula = employed ~ ., family = binomial, data = education_tree_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.9846252   0.1816047  -5.422  5.9e-08 ***
## HighestEducation    0.0108578   0.0046290   2.346  0.018998 *
## GradeCompleted     0.0112573   0.0108677   1.036  0.300272
## AttendedCollege    0.3564761   0.0342495  10.408 < 2e-16 ***
## CurrentCollege     0.4818661   0.0386155  12.479 < 2e-16 ***
## EducationAspiration -0.0096882   0.0036026  -2.689  0.007162 **
## EducationExpectation -0.0728793   0.0204476  -3.564  0.000365 ***
## EducationHours     -0.0006538   0.0001731  -3.777  0.000159 ***
## MotherEducation    -0.0034145   0.0028676  -1.191  0.233763
## FatherEducation    -0.0035278   0.0012059  -2.925  0.003440 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 3324.3  on 2433  degrees of freedom
## Residual deviance: 3046.3  on 2424  degrees of freedom
## AIC: 3066.3
##
## Number of Fisher Scoring iterations: 4

#Tree
set.seed(123)

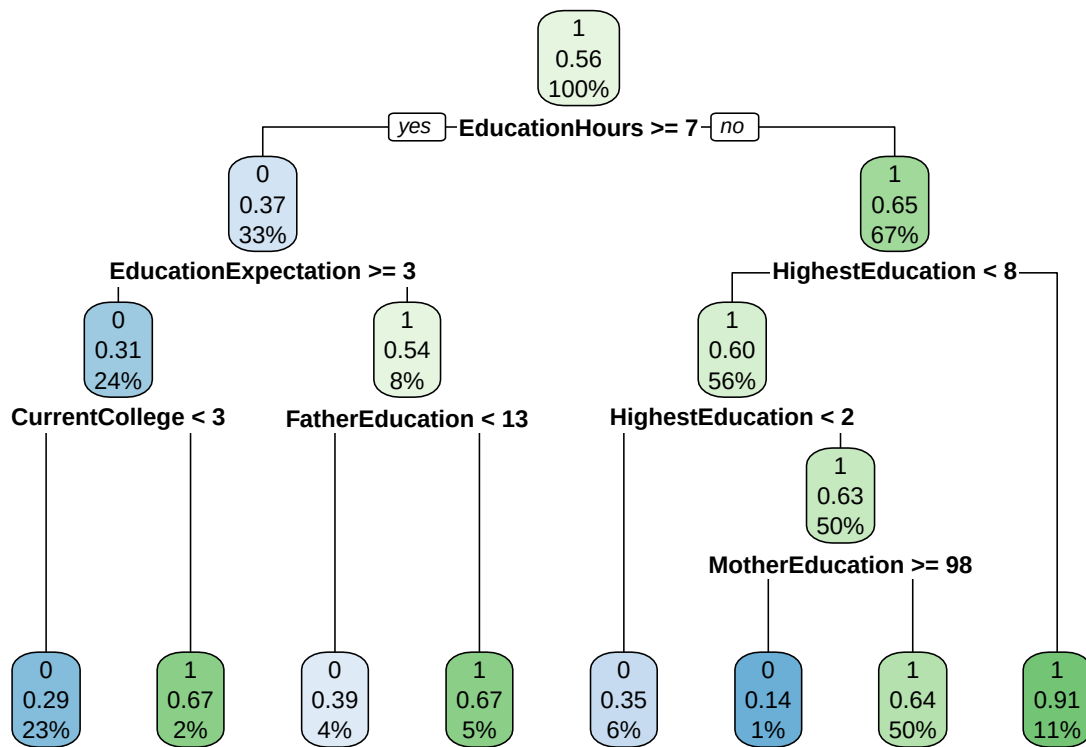
n2 <- nrow(education_tree_data)
train_idx2 <- sample(seq_len(n2), size = 0.7 * n2)

edu_train <- education_tree_data[train_idx2, ]
edu_test  <- education_tree_data[-train_idx2, ]

education_tree <- rpart(
  employed ~ .,
  data = edu_train,
  method = "class"
)

rpart.plot(education_tree)

```



```
edu_pred <- predict(education_tree, edu_test, type = "class")
edu_test_acc <- mean(edu_pred == edu_test$employed)
edu_test_acc
```

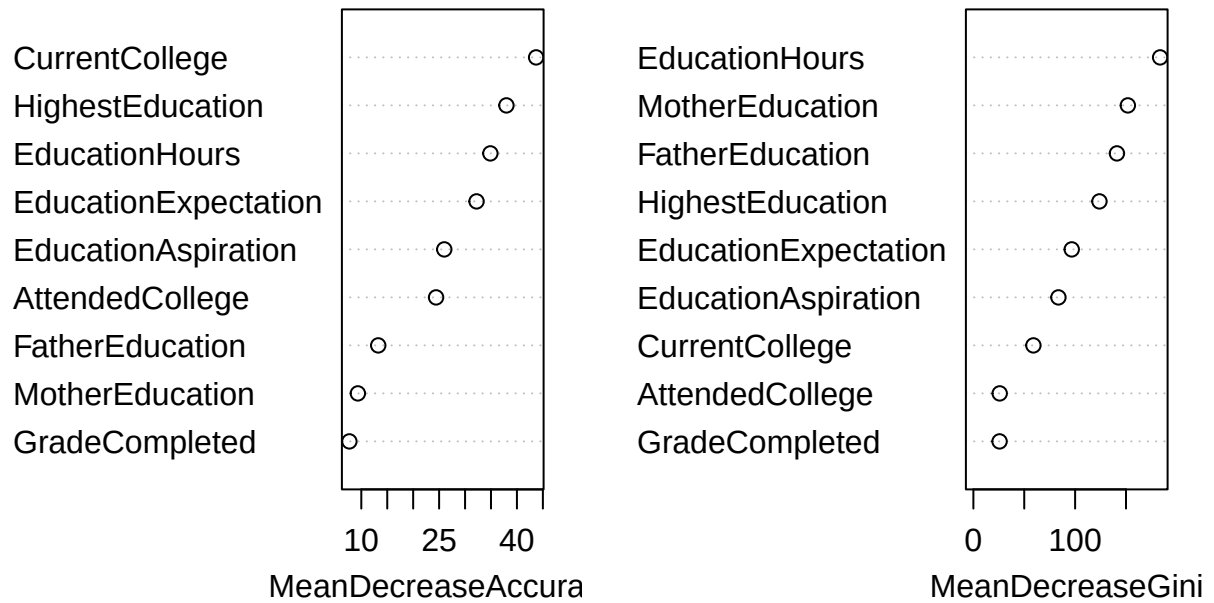
```
## [1] 0.6922025
```

```
# Random Forest
set.seed(123)

education_rf <- randomForest(
  as.factor(employed) ~ .,
  data = education_tree_data,
  ntree = 500,
  importance = TRUE
)

varImpPlot(education_rf)
```

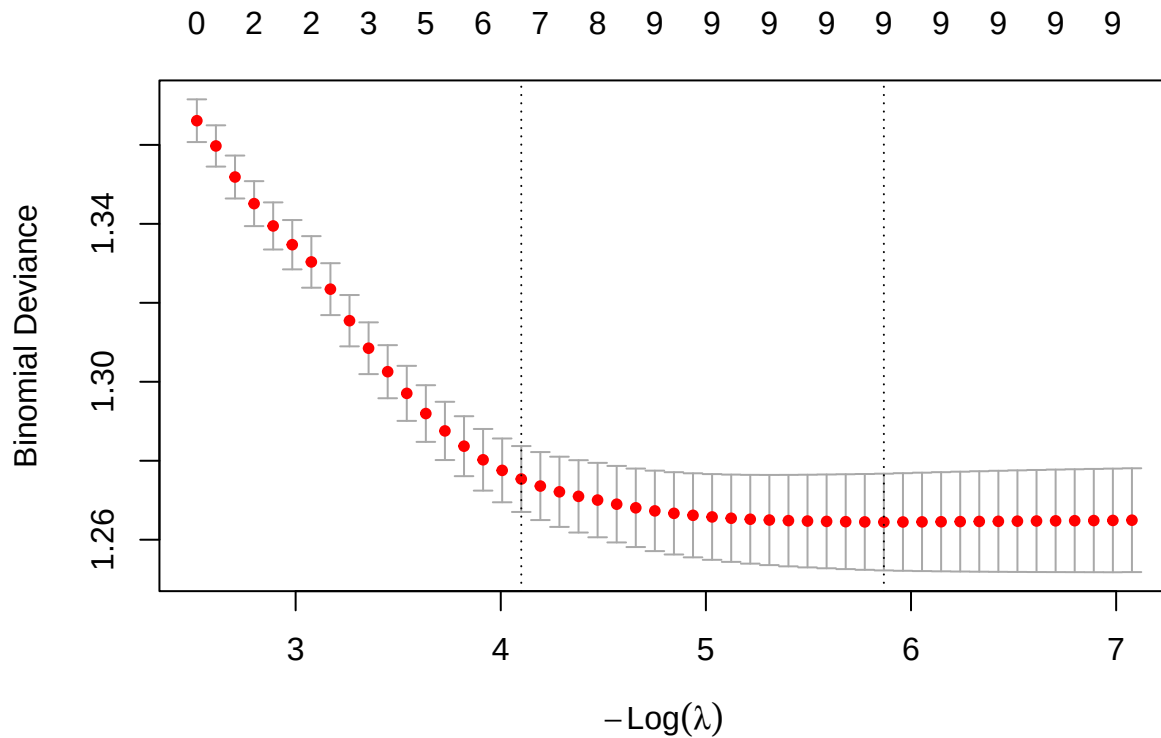
education_rf



```
# LASSO
x_education <- model.matrix(employed ~ ., education_tree_data)[,-1]
y_education <- education_tree_data$employed

education_lasso <- cv.glmnet(
  x_education,
  y_education,
  family = "binomial",
  alpha = 1
)

plot(education_lasso)
```



```
coef(education_lasso, s = "lambda.min")
```

```
## 10 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.min
## (Intercept)    -0.8842140325
## HighestEducation  0.0095812879
## GradeCompleted   0.0076598464
## AttendedCollege  0.3262841722
## CurrentCollege   0.4498574874
## EducationAspiration -0.0079267475
## EducationExpectation -0.0739126671
## EducationHours   -0.0005955542
## MotherEducation  -0.0025398911
## FatherEducation  -0.0030499747
```

```
coef(education_lasso, s = "lambda.1se")
```

```
## 10 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.1se
## (Intercept)    -0.4212732199
## HighestEducation  0.0038430503
## GradeCompleted   .
## AttendedCollege  0.1926939205
## CurrentCollege   0.3080455834
## EducationAspiration -0.0001424525
```

```
## EducationExpectation -0.0794257324
## EducationHours      -0.0003334004
## MotherEducation      .
## FatherEducation      -0.0009165377
```

```
# Final comparison table
```

```
# Final comparison table
```

```
wellbeing_AIC <- AIC(wellbeing_logit)
education_AIC <- AIC(education_logit)

wellbeing_tree_splits <- wellbeing_tree$cptable[nrow(wellbeing_tree$cptable), "nsplit"]
education_tree_splits <- education_tree$cptable[nrow(education_tree$cptable), "nsplit"]

wellbeing_lasso_1se <- coef(wellbeing_lasso, s = "lambda.1se")
education_lasso_1se <- coef(education_lasso, s = "lambda.1se")

wellbeing_lasso_vars <- sum(wellbeing_lasso_1se[-1, ] != 0)
education_lasso_vars <- sum(education_lasso_1se[-1, ] != 0)

wellbeing_rf_acc <- mean(
  predict(wellbeing_rf, wellbeing_tree_data) ==
    wellbeing_tree_data$employed
)

education_rf_acc <- mean(
  predict(education_rf, education_tree_data) ==
    education_tree_data$employed
)

comparison_table <- data.frame(
  Model_Group = c("Well-being", "Education"),
  Logit_AIC = c(wellbeing_AIC, education_AIC),
  Tree_Splits = c(wellbeing_tree_splits, education_tree_splits),
  LASSO_lambda_1se_Selected_Variables =
    c(wellbeing_lasso_vars, education_lasso_vars),
  Random_Forest_Training_Accuracy =
    c(wellbeing_rf_acc, education_rf_acc)
)

comparison_table
```

```
##   Model_Group Logit_AIC Tree_Splits LASSO_lambda_1se_Selected_Variables
## 1 Well-being  3319.918           3                                0
## 2 Education  3066.293           7                                7
##   Random_Forest_Training_Accuracy
## 1                        0.9545075
## 2                        0.9182416
```

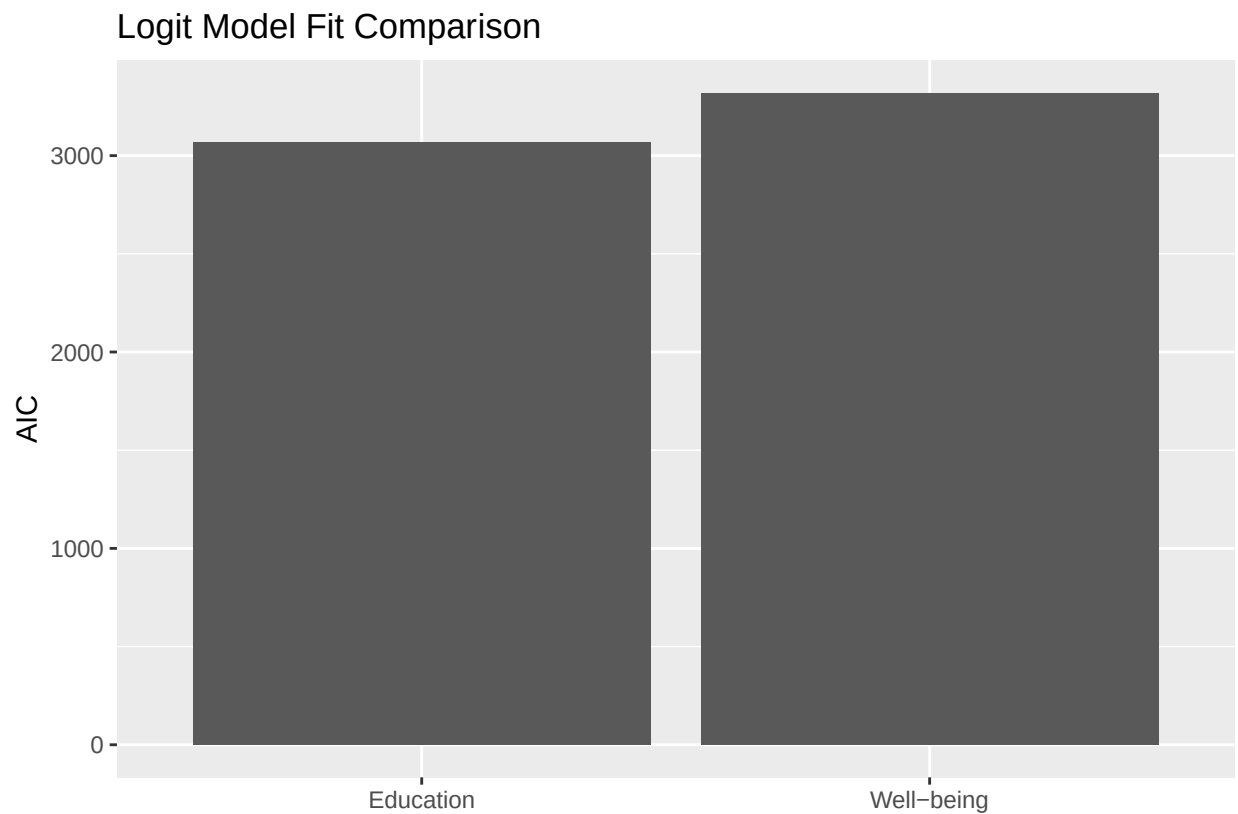
```
ggplot(
  comparison_table,
  aes(
```



```

    x = Model_Group,
    y = Logit_AIC
  )
) +
  geom_col() +
  labs(
    title = "Logit Model Fit Comparison",
    x = "",
    y = "AIC"
  )
)

```

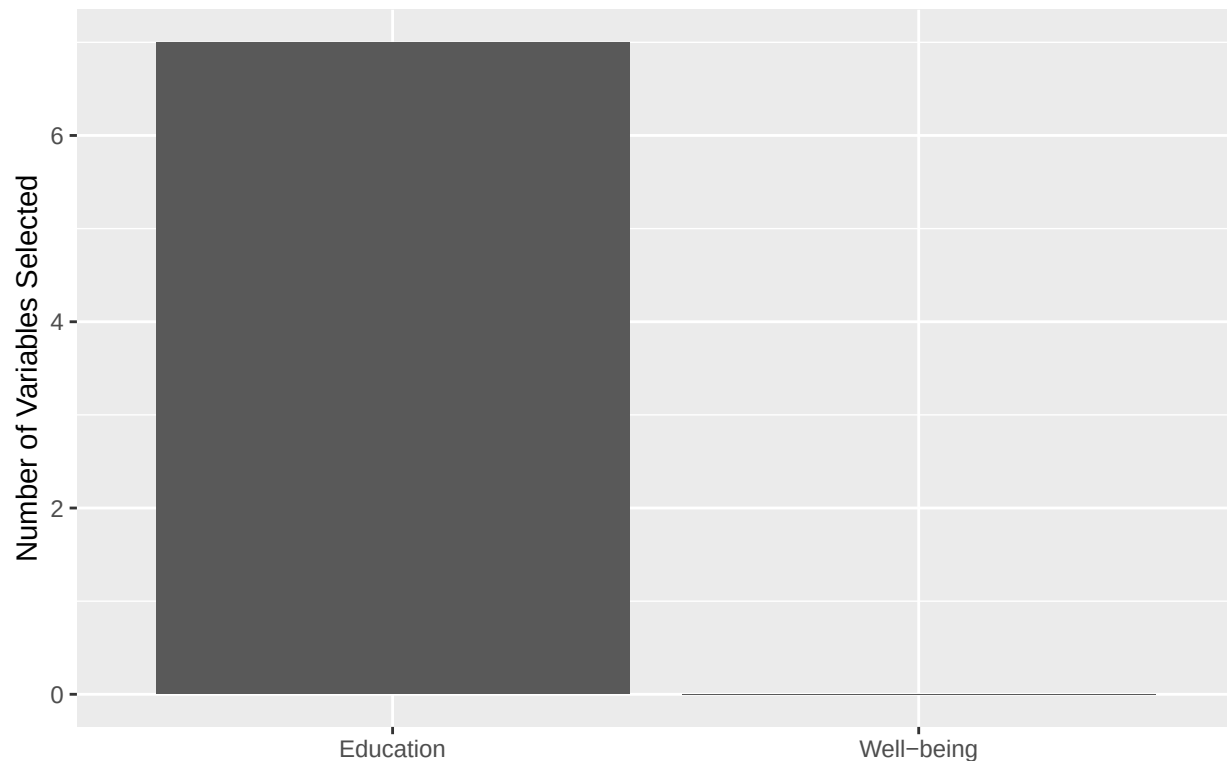


```

ggplot(
  comparison_table,
  aes(
    x = Model_Group,
    y = LASSO_lambda_1se_Selected_Variables
  )
) +
  geom_col() +
  labs(
    title = "LASSO Variable Selection Comparison",
    x = "",
    y = "Number of Variables Selected"
  )
)

```

LASSO Variable Selection Comparison



Overall, the education variables provide stronger and more stable predictive information than the well-being variables. The education logit model has a much lower AIC, the education tree identifies meaningful splits, and LASSO retains several education variables under the conservative λ_{1se} rule. In contrast, the well-being tree does not split and LASSO removes all well-being variables at λ_{1se} . Although the random forest can fit both sets of variables, the broader pattern suggests that education is the stronger predictor of employment outcomes.

```
## Combined Model: Does Well-being Add Anything to Education?
# One common sample containing both the well-being score and the education vars
combined_data <- project_data %>%
  select(employed, wellbeing_score, all_of(education_vars)) %>%
  na.omit()

colnames(combined_data) <- c("employed", "WellbeingScore", education_labels)

cat("Combined-model sample size:", nrow(combined_data), "person-records\n")
```

```
## Combined-model sample size: 2430 person-records
```

```
# Education only (estimated on the combined sample)
combined_edu_only <- glm(
  employed ~ . - WellbeingScore,
  data = combined_data,
  family = binomial
)
```

```
# Education + well-being
```

```
combined_full <- glm(
  employed ~ .,
  data = combined_data,
  family = binomial
)
```

```
summary(combined_full)
```

```
##
## Call:
## glm(formula = employed ~ ., family = binomial, data = combined_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.4001907   0.2567853  -5.453 4.96e-08 ***
## WellbeingScore    0.0797871   0.0411650   1.938 0.052596 .
## HighestEducation  0.0107252   0.0047879   2.240 0.025086 *
## GradeCompleted    0.0214146   0.0111543   1.920 0.054877 .
## AttendedCollege   0.3737597   0.0346282  10.794 < 2e-16 ***
## CurrentCollege    0.4922759   0.0388493  12.671 < 2e-16 ***
## EducationAspiration -0.0072216   0.0036341  -1.987 0.046903 *
## EducationExpectation -0.0700343   0.0205604  -3.406 0.000659 ***
## EducationHours    -0.0006894   0.0001743  -3.956 7.62e-05 ***
## MotherEducation   -0.0035412   0.0028734  -1.232 0.217793
## FatherEducation   -0.0035767   0.0012112  -2.953 0.003147 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 3317.5  on 2429  degrees of freedom
## Residual deviance: 3029.7  on 2419  degrees of freedom
## AIC: 3051.7
##
## Number of Fisher Scoring iterations: 4
```

```
# Does adding well-being improve fit? Likelihood-ratio test, same sample.
```

```
anova(combined_edu_only, combined_full, test = "LRT")
```

```
## Analysis of Deviance Table
```

```
##
```

```
## Model 1: employed ~ (WellbeingScore + HighestEducation + GradeCompleted +
##   AttendedCollege + CurrentCollege + EducationAspiration +
##   EducationExpectation + EducationHours + MotherEducation +
##   FatherEducation) - WellbeingScore
```

```
## Model 2: employed ~ WellbeingScore + HighestEducation + GradeCompleted +
##   AttendedCollege + CurrentCollege + EducationAspiration +
##   EducationExpectation + EducationHours + MotherEducation +
##   FatherEducation
```

```
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
```

```
## 1      2420      3033.5
```

```
## 2      2419      3029.7  1   3.7614 0.05245 .
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# AIC: lower is better. If education-only is not worse, well-being adds nothing.
AIC(combined_edu_only, combined_full)

##           df      AIC
## combined_edu_only 10 3053.492
## combined_full      11 3051.731

# Well-being coefficient before vs after controlling for education
wb_alone    <- summary(wellbeing_logit)$coefficients["wellbeing_score", ]
wb_combined <- summary(combined_full)$coefficients["WellbeingScore", ]

wb_compare <- data.frame(
  Model = c("Well-being only", "Education + Well-being"),
  Estimate = c(
    as.numeric(wb_alone["Estimate"]),
    as.numeric(wb_combined["Estimate"])
  ),
  SE = c(
    as.numeric(wb_alone["Std. Error"]),
    as.numeric(wb_combined["Std. Error"])
  )
)

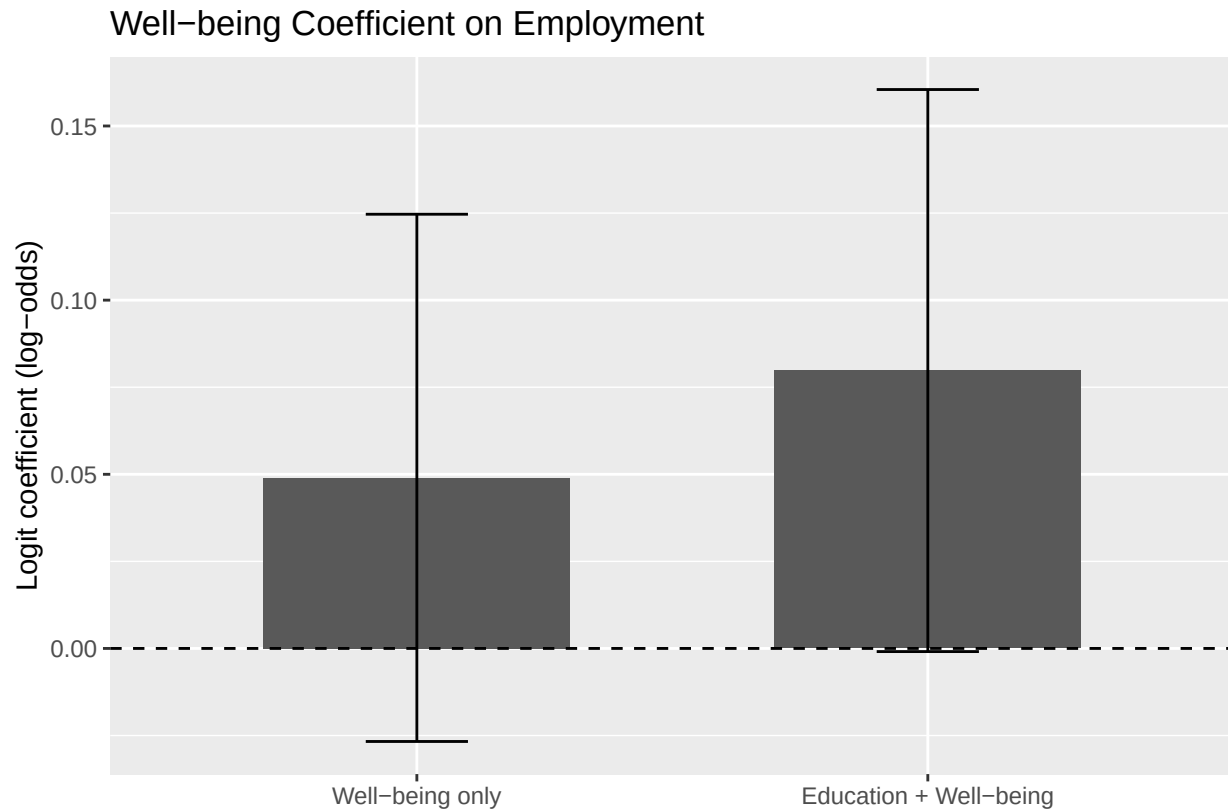
wb_compare$Model <- factor(
  wb_compare$Model,
  levels = c("Well-being only", "Education + Well-being")
)

ggplot(
  wb_compare,
  aes(
    x = Model,
    y = Estimate
  )
) +
  geom_col(
    width = 0.6
  ) +
  geom_errorbar(
    aes(
      ymin = Estimate - 1.96 * SE,
      ymax = Estimate + 1.96 * SE
    ),
    width = 0.2
  ) +
  geom_hline(
    yintercept = 0,
    linetype = "dashed"
  ) +
  labs(
```

```

title = "Well-being Coefficient on Employment",
x = "",
y = "Logit coefficient (log-odds)"
)

```



General idea is that well being coeff actually grows when we mix the sample

```

# Pull Images

library(ggplot2)

# 1. Well-being distribution (ggplot)
ggsave(
  "wellbeing_distribution.png",
  ggplot(project_data, aes(x = wellbeing_score)) +
    geom_histogram(bins = 20) +
    labs(title = "Distribution of Well-being Scores"),
  width = 6, height = 4, dpi = 300
)

```

```

## Warning: Removed 4 rows containing non-finite outside the scale range
## ('stat_bin()').

```

```

# 2. Education tree
png("education_tree.png", width = 2000, height = 1300, res = 200)
rpart.plot(education_tree)
dev.off()

```

```

## pdf
## 2

```

```

# 3. Logit AIC comparison
ggsave(
  "logit_aic_comparison.png",
  ggplot(comparison_table, aes(x = Model_Group, y = Logit_AIC)) +
    geom_col() +
    labs(title = "Logit Model Fit Comparison", x = "", y = "AIC"),
  width = 6, height = 4, dpi = 300
)

# 4. Well-being coefficient comparison
ggsave(
  "wellbeing_coef_comparison.png",
  ggplot(wb_compare, aes(x = Model, y = Estimate)) +
    geom_col(width = 0.6) +
    geom_errorbar(
      aes(ymin = Estimate - 1.96 * SE, ymax = Estimate + 1.96 * SE),
      width = 0.2
    ) +
    geom_hline(yintercept = 0, linetype = "dashed") +
    labs(title = "Well-being Coefficient on Employment",
         x = "", y = "Logit coefficient (log-odds)",
         width = 6, height = 4, dpi = 300
    )
)

```